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Close-pack, high-aspect-ratio camera tripod

Abstract

Tripods inclusive of a hub defining a center bore; a set of legs, each leg in the set of legs pivotably or retractably coupled to the hub; a center column comprising a first non-circular cross-section configured to translate within the center bore; and a head coupled to the center column configured to couple to a camera are disclosed.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This Application is a continuation application of U.S. patent application Ser. No. 17/708,962, filed on 30 Mar. 2022, which is a continuation application of U.S. patent application Ser. No. 17/127,944, filed on 18 Dec. 2020, issued as U.S. Pat. No. 11,320,724 on 3 May 2022, which is a continuation application of U.S. patent application Ser. No. 15/931,503, filed on 13 May 2020, issued as U.S. Pat. No. 10,901,301 on 26 Jan. 2021, which claims the benefit of U.S. Provisional Application No. 62/965,597, filed on 24 Jan. 2020 and is a continuation-in-part application of U.S. patent application Ser. No. 16/501,118, filed on 13 May 2019, issued as U.S. Pat. No. 10,982,806 on 20 Apr. 2021, each of which is incorporated in its entirety by this reference.

TECHNICAL FIELD

(1) This invention relates generally to the field of photography and more specifically to a new and useful close-pack, high-aspect-ratio camera tripod in the field of photography.

Description

BRIEF DESCRIPTION OF THE FIGURES

- (1) FIGS. 1-5 are schematic representations of a tripod;
- (2) FIG. 6 is a schematic representation of a hanging hook and a mobile mount;
- (3) FIGS. 7A and 7B are schematic representations of a mobile mount;
- (4) FIGS. 8A and 8B are schematic representations of leg clamps;
- (5) FIGS. 9A and 9B are schematic representations of a leg assembly;
- (6) FIG. 10 is a schematic representation of a hub;
- (7) FIG. 11 is a schematic representation of the tripod; and
- (8) FIG. 12 is a schematic representation of a camera locking tab.

DESCRIPTION OF THE EMBODIMENTS

(9) The following description of the embodiments of the invention is not intended to limit the invention to these embodiments but rather to enable a person skilled in the art to make and use this invention.

1. Tripod

(10) As shown in FIGS. 1-5, a tripod **100** includes: a hub **140** defining a center bore **142** and a set of leg mounts **144** arranged in a radial pattern about the center bore **142**; a set of legs **160**, each leg

160 in the set of legs **160** pivotably coupled to a leg mount **144** in the set of leg mounts **144** and configured to telescopically extend from the hub **140**; a center column **150** configured to translate within the center bore **142** of the hub **140** and including a spherical end **156** configured to nest between the leg mounts **144**. The tripod further includes a head **110** pivotably coupled to the spherical end **156** and including: a base section **112**; a camera platform **130** arranged over the base section **112**, defining a rail **134** and a locking tab **132**, and configured to transiently receive a camera adapter coupled to a camera; a set of flanges **114** arranged in the radial pattern, extending below the base section **112** opposite the camera platform **130**, extending around a section of the spherical end **156**, and configured to nest between the leg mounts **144**; a hat **116** arranged in the base section **112** over the spherical end **156**; a pivot control ring **124** arranged about the base section **112**, configured to drive the hat **116** into the spherical end **156** to fix an orientation of the head **110** on the spherical end **156** responsive to rotation in a first direction about the base section **112**, and configured to retract the hat **116** from the spherical end **156** to unlock the head **110** from the spherical end **156** responsive to rotation in a second direction about the base section **112**.

(11) In one variation, the tripod further includes a camera lock ring **126** arranged proximal the pivot control ring **124**, concentric with the pivot control ring **124**, and configured to drive the camera locking tab **132** toward the rail **134** to transiently lock the camera adapter between the camera locking tab **132** and the rail **134**.

(12) The tripod **100** includes a set of legs **160**. Each leg **160** of the tripod **100** can include a series of nesting telescopic leg segments **162** wherein each leg segment—other than the first, largest leg segment—in a leg is configured to nest within an adjacent, larger leg segment of larger cross-section. Furthermore, the distal end of each leg segment—other than the last, smallest leg segment—in a leg can include a clamp assembly configured to selectively clamp an adjacent, smaller leg segment **162**, thereby enabling this smaller leg segment **162** to telescope within the adjacent, larger leg segment **162**. Each leg connects to the hub **140** via a leg mount **144** including a multi-stage leg position stop.

(13) In one variation, the tripod **100** includes a spherical end **156** and a head **110** pivotably coupled to the spherical end **156**, the head **110** including: a base section **112**; a camera platform **130** arranged over the base section **112** and configured to transiently receive a camera adapter coupled to a camera; a threaded section **118** extending along a central axis of the head **110** and arranged over the spherical end **156**; a sun gear **120** threaded onto the threaded section **118** and configured to translate along the threaded section **118** when rotated; a hat **116** arranged on to the sun gear **120** and facing the spherical end **156**; a spring **117** arranged between the hat **116** and the sun gear **120** and configured to depress the hat **116** against the spherical end **156**; a set of planet gears **122** arranged about and meshed with the sun gear **120**; and a set of flanges **114** extending from the base section **112** opposite the camera platform **130**, extending around a section of the spherical end **156**, and arranged in a radial pattern about the spherical end **156**. In this variation, the tripod **100** further includes a pivot control ring **124** arranged about the base section **112**, comprising a ring gear meshed with the set of planet gears **122**, and configured to: rotate the sun gear **120** about the threaded section **118** via the set of planet gears **122**, drive the hat **116** toward the spherical end **156**, clamp the spherical end **156** against the set of flanges **114**, and fix an orientation of the head **110** on the spherical end **156** responsive to rotation in a first direction about the base section **112**; and rotate the sun gear **120** about the threaded section **118** via the set of planet gears **122**, retract the hat **116** from the spherical end **156**, and unlock the head **110** from the spherical end **156** responsive to rotation in a second direction opposite the first direction.

2. Applications

(14) Generally, the tripod **100** includes: a hub **140** defining a set of leg mounts **144** pivotably coupled to a set of legs **160**; and a head **110** including a set of stacked control rings that enable a user to rapidly adjust pitch, yaw, and roll of a camera—mounted to the head **110**—relative to the hub and legs and to rapidly install, lock, and remove the camera with a single hand in the same

location. More specifically, the tripod **100** includes a set of stacked control rings that fall to hand in one compact location and thus enable a user to manipulate the position of a camera, and quickly mount and dismount the camera from the tripod **100**. For example, the tripod can include a set of concentric control rings stacked just below the camera mount which may be fully engaged and disengaged with less than one full turn (e.g., less than 360-degrees), thereby allowing a user to quickly and easily move the head **110** a full 360-degrees in pan, easily move the head **110** nearly a full 180-degrees in tilt (e.g., pitch and roll), and then fully and confidently lock the tripod **100** in place without repositioning her hand or removing her hand from the head **110**.

(15) Furthermore, by including the set of stacked, concentric control rings, the tripod **100** condenses the pan, tilt, and lock/unlock controls in one compact location, thereby: limiting features projecting outwardly from the head **110**; minimizing effective diameter of the head **110**; increasing compactness and space efficiency of the tripod **100** when fully collapsed; reducing weight of the tripod **100**; and improving ease of transport, storage, and accessibility of the tripod **100** for a user.

(16) The head **110** of the tripod **100** is mounted to a center column **150** configured to run inside a center bore **142** of the hub **140**, and the hub **140** defines a set of leg mounts **144** that couple and support a set of legs **160**. The center column **150** defines a spherical end **156**, and the head **110** defines a set of flanges **114** extending from the bottom of the head **110** to form a socket around the spherical end **156**, which enables a user to tilt the head **110** relative to the hub **140**. In particular, the head **110** defines a set of flanges **114** arranged in a radial pattern matched to a radial pattern of leg mounts **144** extending from the hub **140** such that—when the tripod **100** is fully collapsed—the head **110** is radially offset (e.g., by 60-degrees) from the hub **140** to enable the flanges **114** and the leg mounts **144** to nest (or “interlock”), to encapsulate the spherical end **156**, and to thus achieve high vertical and volumetric packing efficiency. The interlocking head **110**, hub **140**, and leg sections form a solid and robust collapsed state—such that the tripod **100** maintains a substantially uniform effective diameter when collapsed—which enables the user to pack away the tripod **100** without extraneous knobs or protrusions snagging on other equipment or bag flaps/openings. For example, when fully collapsed (e.g., in the collapsed position), the tripod **100** can approximate a cylindrical form with minimal negative space, thereby exhibiting high volumetric efficiency. Furthermore, in this example, the center column **150** can define a triangular section such that—when the center column **150** is fully retracted from the hub **140** with the head **110** nested around the leg mounts **144**—the interior faces of the legs **160** mate (or fall very near) the exterior faces of the center column **150**, thereby minimizing negative space inside the cylindrical exterior form approximated by the tripod **100** in this collapsed state.

(17) In one variation, the radial distance between flanges **114** can be less than the radial width of the center column **150** such that the head **110** can tilt nearly 180-degrees about the spherical end **156** in both pitch and yaw directions. For example, a user may: shoot a first series of photos with her camera—installed on the head **110**—retained in a landscape position by the head **110**; and then manipulate the pivot control ring **124** to rapidly unlock, tilt, and relock the head **110** to relocate the camera in a portrait position. The user may also manipulate the pivot control ring **124** to loosen the head **110** on the spherical mount in order to enable more subtle pitch adjustments of the camera in this portrait position, such as within a range of 120-degrees less a sum of radial widths of the center column **150** and one flange.

(18) Each leg **160** of the tripod **100** includes a set of nested leg segments **162** (or “telescoping stages”), and the tripod **100** also includes a center column **150**, all of which cooperate to enable the tripod **100** to expand to a height several times (e.g., four times) the height of the tripod **100** in the collapsed state. When opened, the tripod **100** can occupy a range of footprints and heights, thereby defining a robust structure for support of heavy camera equipment (e.g., sandbags, telephoto lenses, etc.) and supporting a wide range of applications and uses for a photographer.

3. Head

(19) As shown in FIGS. 1-3, the head **110** includes: a camera platform **130** arranged orthogonally

to the central axis of the head **110**; a camera lock ring radially operated about the central-axis of the head **110** to interface with a locking tab **132**; a flanged socket—including a set of (e.g., three) flanges arranged in a radial pattern—configured to receive the spherical end **156** on the center column **150**; a hat **116** interposed between the camera platform **130** and the spherical end **156** and configured to cooperate with the set of flanges to clutch the spherical end; and a pivot control ring **124** radially-operable (i.e., rotatable) about the central-axis of the head **110** to drive the hat **116** into and away from the spherical end **156** in order to lock and release the head **110** from the spherical end **156**, respectively.

(20) In one implementation, the head further includes: a threaded section **118** (e.g., a threaded bore or a threaded shaft) extending along a central axis of the head **110** and arranged over the spherical end **156**; a sun gear **120** threaded onto the threaded section **118** and configured to translate along the threaded section **118** when rotated; and a set of planet gears **122** arranged about and meshed with the sun gear **120**. The pivot control ring **124** can also include a ring gear mated to the set of planet gears **122**; and the hat **116** can be mounted to the sun gear **120**. The head **110** can also include a spring **117** arranged between the hat **116** and the sun gear **120** opposite the spherical end **156** and configured to bias the hat **116** toward the spherical end **156** and to clutch the spherical end **156** between the hat **116** and the set of flanges **114**, thereby limiting rotation of the head **110** on the spherical end **156** even when the pivot control ring **124** is unlocked with the sun gear **120** spun up the threaded section **118** and retracted from the spherical end **156**. Therefore, rotation of the pivot control ring **124** about the head **110** in a first direction can rotate the set of planet gears **122** in a second direction and thus rotate the sun gear **120** in the first direction, thereby: spinning the sun gear **120** down the threaded section **118**; compressing the spring **117** between the sun gear **120** and the hat **116**; positively engaging a leading face of the sun gear **120** against a back face of the hat **116**; and thus driving the hat **116** against the spherical end **156** to lock the spherical end **156** between the hat **116** and the set of flanges **114**—thereby locking the pitch, yaw, and roll position of the head **110** on the spherical end **156**. Similarly, rotation of the pivot control ring **124** about the head **110** in the second direction can rotate the set of planet gears **122** in the first direction and thus rotate the sun gear **120** in the second direction, thereby: spinning the sun gear **120** up the threaded section **118**; retracting the leading face of the sun gear **120** from the back face of the hat **116**; (partially) releasing the spring **117**; and thus reducing compression of the spherical end **156** between the hat **116** and the set of flanges **114**—thereby unlocking the head **110** from the spherical end **156**.

(21) Therefore, the threaded section **118**, the sun gear **120**, and the set of planet gears **122** can cooperate with the pivot control ring **124**, the hat **116**, and the spherical end **156** to lock and unlock an orientation of the head **110** about the spherical end **156**.

(22) The camera platform **130** includes a substantially planar top surface configured to receive the base or side of a camera, camera mount, or adapter. The camera platform **130** also includes a projected fixed rail **134** to mate against a side of a camera, camera mount, or adapter. The operable locking tab **132** cooperates with the fixed rail **134** to locate and retain a camera adapter mounted to a camera in order to restrict movement of the camera relative to the head **110**. Furthermore, the head **110** can include a spring **136** that biases the camera locking tab **132** toward the fixed rail **134** in order to snap the camera adapter onto the camera platform **130** when the camera is offered up to the head **110**. Moreover, the camera lock ring **126** can define a ramp or cam that drives and retains the camera locking tab **132** toward the fixed rail **134** in order to lock the camera adapter between the camera locking tab **132** and the camera fixed rail **134**. The camera lock ring **126** slides around the central axis of the head **110**. Therefore, the fixed rail **134**, the camera locking tab **132**, the spring **136**, and the camera lock ring **126** can cooperate to enable a user to drop the camera onto the head **110** with her left hand and then—while the spring **136** drives the camera locking tab **132** against the camera adapter to loosely retain the camera on the camera platform **130**—rotate the camera lock ring **126** with her left hand to fully lock the camera to the head **110** (e.g., all while

reaching for a lens in her camera bag with her right hand).

(23) Then, the user may slip her left hand down (e.g., by approximately 10 millimeters) to locate her fingers off of the camera lock ring **126** and onto the pivot control ring **124**, rotate the pivot control ring **124** to loosen the head **110** on the spherical end **156**, and adjust the tilt and pan of the head **110**—and therefore the camera—relative to the hub **140** to locate a target scene in the field of view of the camera before retightening the pivot control ring **124**. The user may begin shooting the target scene immediately thereafter.

(24) Furthermore, the user may keep her left hand on the head **110** (with her fingers in contact with the pivot control ring **124**) in order to make on-the-fly pan and tilt adjustments to the camera by loosening the pivot control ring **124** with her left hand, repositioning the head **110** with her left hand, and then retightening the pivot control ring **124** again with her left hand before resuming shooting.

(25) Finally, the user may raise her left hand up the head **110** to engage the camera lock ring **126** and rotate the camera lock ring **126** to release the camera locking tab **132**; the spring **136** can continue to bias the camera locking tab **132** toward the fixed rail **134** in order to retain the camera on the head **110** until the user biases the camera locking tab **132** (e.g., with her left hand) to retrieve the camera.

(26) Therefore, the head **110** can define a compact set of stacked controls that enable a user to rapidly and easily install, adjust, and remove a camera from the tripod **100** with a single hand.

(27) 3.1 Camera Platform

(28) The camera platform **130** can include a top section that defines: a camera platform **130** (e.g., a grooved or textured surface) configured to carry a vertical load of a camera, camera mount, or other adapter; a fixed rail **134** extending along a first end of the camera platform **130** surface and defining an undercut section **190**; and a pass-through for the camera locking tab **132** at a second end of the camera mount surface. The camera platform **130** can also include a spring **136** that biases the camera locking tab **132** toward the fixed rail **134** and that enables retraction of the camera locking tab **132** when a camera mount is installed over the camera mount surface. The camera locking tab **132** can similarly define an undercut section **192** and can cooperate with the fixed rail **134** to transiently receive and retain a camera, a camera mount, or other adapter over the camera mount surface. The camera platform **130** also includes a bottom section that defines: a threaded section **118** (or threaded shaft) configured to mate with the threaded end of the sun gear **120**; and a bore for a spring **117** and a detent-pin configured to engage detent surfaces (e.g., ridges) along the adjacent camera lock ring **126**.

(29) In one implementation, the camera platform **130** can be manufactured (e.g., cast, machines) from aluminum, steel, or a rigid polymer.

(30) 3.2 Camera Lock Ring

(31) In one implementation, as shown in FIG. **11**, the head **110** includes a camera lock ring **126** arranged proximal and concentric with the pivot control ring **124** and configured to drive the camera locking tab **132** toward the fixed rail **134** to transiently lock the camera adapter between the camera locking tab **132** and the fixed rail **134**.

(32) In one implementation, the camera lock ring **126** includes an annular ring including: on a first face, a set of detents on a first side configured to interface with a spring-loaded detent-pin; and on a second side, a ramp configured to interface with the camera locking tab **132** such that rotating the annular ring about the central axis of the head **110** forces the camera locking tab **132** into a series of locked positions along the ramp, which dynamically fastens the camera, camera mount, or camera adapter to the camera platform **130**. In this implementation, the camera locking tab **132** can be configured to actuate into a recess of the camera, camera mount, and/or camera adapter to restrict movement of the camera responsive to rotation of the annular ring. The camera locking tab **132** can be spring-loaded and interface with a ramp on a ring (e.g., camera lock ring **126**) parallel to the camera platform **130**. As the annular ring rotates about the central axis of the head **110**, the ramp

can force the tab into a fixed position, locking the camera (or camera accessory) to the camera platform **130**. A series of detents sit opposite the ramp on the ring and interface with a spring-loaded detent-pin to stagger the locking positions of the tab.

(33) In one variation, the camera lock ring **126** includes a protrusion extending radially outward (e.g., a finger tab) to interface with a user's finger during one-handed manipulation. The annular ring can sit in a plane parallel to the camera platform **130**. In one variation, the ring can be arranged immediately below the camera platform **130**. For example, the annular ring can be manufactured in aluminum, plastic, or carbon fiber.

(34) Therefore, when the camera lock ring **126** is unlocked, the camera locking tab **132** can pivot or slide within the camera platform **130** to engage and retain a camera mount, thus enabling a user to locate one end of the camera mount within the undercut section **190** of the fixed rail **134**, rest the opposing end over the camera locking tab **132**, and press down. When the user rotates the camera lock ring **126** to the lock position, a cam surface defined by the camera lock ring **126** closes toward the camera locking tab **132** and then engages the camera locking tab **132** to prevent retraction of the camera locking tab **132** from the rail **134** responsive to the user pushing or pulling on the camera locking tab **132**, thereby positively locking the camera mount between the camera locking tab **132** and the rail **134**.

(35) More specifically, the camera lock ring **126** can overdrive the camera locking tab **132** toward the fixed rail **134** and thus function as an ancillary lock for the camera platform **130**.

(36) 3.3 Locking Tab

(37) In one implementation, the camera locking tab **132** and the spring **136** cooperate to retain a camera mount—affixed to a camera—on the camera platform **130** without further positive lock by the camera lock ring **126** such that the user may push, pull, and/or pivot the camera without the camera locking tab **132** releasing the camera mount from the camera platform **130**.

(38) In one implementation shown in FIG. **12**, the undercut section **190** of the fixed rail **134** can be configured to mate with a first beveled face of a camera mount (or a “camera adapter”). The undercut section **192** of the camera locking tab **132** can be configured to mate with a second beveled face of the camera mount opposite the first beveled face. For example, the undercut sections **190**, **192** of the fixed rail **134** and the camera locking tab **132** can define complementary 45° beveled faces when assembled over the camera platform **130**. In this implementation, the camera locking tab **132** is mounted to and pivots about a pivot **194** (e.g., a pin) arranged under the camera platform **130**. The spring **136** is laterally offset from the pivot **194** and drives (e.g., pivots) the camera locking tab **132** upward to mate the undercut section **192** against the second beveled face of the camera mount and thus retain the camera mount on the camera platform **130**.

(39) In particular, the pivot **194** can be located along (or near) a vector that intersects and is normal to the undercut section **192** of the camera locking tab and the second beveled face of the camera mount when assembled over the camera platform **130**. Because the pivot **194** is located along this vector: the effective lever arm length of the camera mount applied to the camera locking tab **132** is null (or nearly null) a lever arm; and the effective torque applied on the camera locking tab **132** by the camera mount—such as when the camera is pulled or rotated over the camera platform **130**—is null (or nearly null) and (nearly) decoupled from the magnitude of the force or torque applied to the camera. Furthermore, because the spring **136** is laterally offset from the pivot **194**, this effective torque applied on the camera locking tab **132** by the camera mount is less than the opposing torque applied to the camera locking tab **132** by the spring **136** such that the camera locking tab **132** remains engaged to the camera mount despite the magnitude of the force or torque applied to the camera. Thus, when a user pushes, pulls, or pivots the camera, the resulting torque to open the camera locking tab **132** is (approximately) null, and the camera locking tab **132** therefore does not rotate away from the camera mount. Therefore, the camera locking tab **132** remains fixed in its closed position and retains the camera mount and the camera in place over the camera platform **130** despite forces applied to the camera.

(40) However, the camera locking tab **132** can pivot about the pivot **194** responsive to a user pushing or pulling downward on the camera locking tab **132** directly, which withdraws the undercut section **192** away from the adjacent second bevel on the camera mount and enabling the user to lift the camera and the camera mount off of the camera platform **130**.

(41) Furthermore, responsive to a downward force on the top of the camera locking tab **132** over the undercut section **192** by the camera mount during installation of the camera mount onto the camera platform **130**, the camera locking tab **132** can pivot downward about the pivot **194**, thereby withdrawing the undercut section **192** away from the camera platform **130** and enabling the camera mount to move downward toward the camera platform **130**. In particular, the user may insert the first beveled face of the camera mount within the undercut section **190** of the fixed rail **134**, rest the second beveled face of the camera mount over the camera locking tab **132**, and press down. The force of the second beveled face of the camera mount on the camera locking tab **132** counters the spring **136** and applies a torque to the camera locking tab **132**, thereby rotating the camera locking tab **132** downward about the pivot **194** to open the camera platform **130** to accept the camera mount. The second beveled face of the camera mount slides along the apex of the camera locking tab **132** over the undercut section **192** as the camera locking tab **132** opens and eventually drops past the apex of locking tab **132** to seat under the camera locking tab **132** with the second beveled face positioned against the undercut section **192** of the camera locking tab **132** and with the base of the camera mount now in contact with the top surface the camera platform **130**. The spring **136** then automatically drives the camera locking tab **132** upward to positively clutch the camera mount between the rail **134** and the camera locking tab **132**.

(42) 3.4 Controls Chassis

(43) A controls chassis: is interposed between the camera platform **130** and the set of flanges **114**; houses the sun gear **120**, planet gears **122**, and hat **116**; and locates the pivot control ring **124** below the camera lock ring **126**. The controls chassis can also define a set of bear surfaces or posts configured to locate the planet gears in a radial pattern about the central axis of the head **110**, as shown in FIG. 2.

(44) 3.4.1 Pivot Control Ring

(45) The tripod includes a pivot control ring **124** arranged about the base section **112** of the head **110** and configured to fix an orientation of the head **110** on the spherical end **156** or unlock the head **110** from the spherical end **156** responsive to rotation by a user.

(46) The pivot control ring can define an outer annular ring, such as including a splined or grooved outside face configured for hand manipulation. The interior face of the pivot control ring **124** can also define an annular ring gear configured to mesh with the set of planet gears **122** arranged within the controls chassis and camera platform **130**.

(47) The pivot control ring can be arranged on the head **110** of the tripod **100** and can be accessible by hand. Rotating the pivot control ring thus rotates the planet gears **122**, which rotates the sun gear **120** about the threaded section **118** within the camera platform **130**, thereby causing the sun gear **120** to translate linearly along the central axis of the head.

(48) 3.4.2 Planetary Gearbox

(49) The tripod **100** also includes a planetary gearbox—including the sun gear **120** and the set of planet gears **122**—arranged in the controls chassis and configured to transform rotation of the pivot control ring **124** into linear movement of the hat **116**.

(50) The sun gear **120** revolves about the central axis of the head **110**. The height of the sun gear **120** can approximate (or exceed) the sum of: the height of the planet gears **122**; and the range of vertical motion of the sun gear **120** between the locked and unlocked positions of the pivot control ring **124**. The sun gear **120** includes a coaxial (internal or external) threaded section that mates with (i.e., threads onto) the threaded section **118** in the head such that the sun gear **120** raises and lowers within the head—and thus retracts and advances the hat **116** toward the spherical end **156**—when the pivot control ring **124** is rotated about the head **110**. For example, the threaded section **118**

within the head **110** and the sun gear **120** can define single-lead or double-lead ACME threads, which may limit friction between the threaded section **118** and the sun gear **120** when the sun gear **120** is rotated via the pivot control ring **124**.

(51) Furthermore, each planet gear **122**: can include a shaft or pin extending parallel to the central axis of the head **110** and seated in complementary mounting bores in the controls chassis and the camera platform **130**; and can mesh with both the pivot control ring **124** and the sun gear **120** such that rotation of the pivot control ring **124** rotates the sun gear **120** about the threaded section **118** and raises and lowers the sun gear **120**—and therefore the hat **116**—toward the spherical end **156**. In one variation, the pivot control ring **124** can be arranged immediately below and coaxial with the camera lock ring **126**.

(52) 3.4.3 Friction Hat

(53) In one implementation, the friction hat **116** (hereinafter the “hat”) is contiguous with the sun gear **120**. For example, the sun gear **120** can include a concave spherical cup section coaxial with the threaded section **118**, facing the spherical end **156**, and configured to engage and clutch against the spherical end **156** when actuated by the pivot control ring **124**.

(54) Alternatively, the hat **116** can be distinct from and coupled to the sun gear **120**. For example, the sun gear **120** can define a shoulder (or a bore) coaxial with the threaded section **118**; and the hat **116** can include a complementary feature that mates with, slides along, and rotates about the shoulder (or bore) of the sun gear **120**. In this implementation, the hat **116** can also define a concave spherical cup section coaxial with the threaded section **118**, facing the spherical end **156**, and configured to engage and clutch against the spherical end **156** when actuated by the pivot control ring **124**. The tripod **100** can also include the spring **117** arranged about this shoulder (or within this bore) and configured to bias the rear face of the hat **116** away from the sun gear **120** and toward the spherical end **156**. Alternatively, a set of (e.g., three) counterbores can be arranged in a radial pattern about the sun gear **120** and/or the hat **116**, and a set of springs **117** can be installed in these counterbores to bias the rear face of the hat **116** away from the sun gear **120** and toward the spherical end **156**.

(55) Therefore, the spring(s) **117** can depress the rear face of the hat **116** off of the sun gear **120** and toward the spherical end **156**. When the pivot control ring **124** is rotated toward the lock position, the sun gear **120** can run down the threaded section **118** and drive toward the spherical end **156** such that the shoulder (or the bore) drives into the hat **116**, thereby compressing the spring **117**. In addition, because the hat **116** is radially isolated from the sun gear **120** and biases against the spherical end **156** by the spring **117**, the hat **116** may remain stationary against the spherical end **156** as the sun gear **120** is driven down toward the spherical end **156**, thereby reducing wear on the hat **116** and the spherical end **156**. Further rotation of the pivot control ring **124** drives the leading face of the sun gear **120** into contact with the rear face of the hat **116** and then rigidly locks the hat **116** against the spherical end **156**, thereby rigidly locking the spherical end **156** between the hat **116** and the set of flanges **114**.

(56) More specifically, when the pivot control ring **124** is rotated in a first direction, the ring gear integrated into the pivot control ring **124** rotates the set of planet gears **122**, which in turn rotate the sun gear **120** in the first direction, thereby unthreading the sun gear **120** from the threaded section **118** of the head **110**, driving the hat **116** into the spherical end **156** below, and thus clamping the spherical end **156** against the flanges **114** extending from the base section **112** around the spherical end **156** below. Similarly, when the pivot control ring **124** is rotated in the opposite direction, the ring gear rotates the set of planet gears **122**, which in turn rotate the sun gear **120** in a second direction, thereby threading the sun gear **120** into the threaded section **118** of the head **110**, retracting the hat **116** from the spherical end **156** below, and thus releasing the spherical from the flanges **114** below.

(57) Furthermore, when the sun gear **120** is retracted from the spherical end **156**, the spring **117** can function to drive the hat **116** into the spherical end **156** in order to maintain a minimum amount of

friction between the hat **116** and the spherical end **156**, thereby retaining the orientation of the head **110** relative to the spherical mount and preventing rotation of the head **110** relative to the spherical end **156**, such as when a user rotates the pivot control ring **124** in the first direction to tighten the hat **116** against the spherical end **156**. More specifically, the spring **117** and the hat **116** can cooperate to resist a torque applied to the pivot control ring **124** in order to prevent rotation of the head **110** relative to the spherical end **156** when the pivot control ring **124** is rotated—such as within a single hand—in the first direction to tighten the hat **116** onto the spherical end **156**.

(58) 3.4.4 Panning Control Ring

(59) In one implementation, the head **110** includes a panning control ring **128**. In this implementation, the head defines an upper body coupled to the base section **112** and rotatable about a pan axis of the base section **112**. The panning control ring **128** can be arranged in between the camera lock ring **126** located on the upper section and the pivot control ring **124** located on the base section **112** and configured to lock the upper body to the lower body responsive to rotation in the first direction about the base section **112**. Further, the panning control ring **128** can be configured to unlock the upper body from the base section **112** responsive to rotation in the second direction.

(60) For example, the controls chassis can couple to the camera platform **130** via a radial bearing or bushing and define an upper section containing a second threaded section. The camera platform **130** can include a shoulder adjacent the upper section of the controls chassis, with the panning control ring **128** threaded onto the second threaded section and abutting the shoulder of the controls chassis. In this example, rotation of the panning control ring **128** in a first direction threads the panning control ring **128** downward onto the second threaded section, thereby engaging and constraining the shoulder of the camera platform **130** between the panning control ring **128** and the controls chassis. Rotation of the panning control ring in a second direction unthreads the panning control ring **128** from the second threaded section, thereby releasing the shoulder of the camera platform **130** from between the panning control ring **128** and the controls chassis and enabling the camera platform **130** to rotate—or “pan”—about the controls chassis.

(61) The panning control ring **128** may be operated by a user with one hand by actuating the ring radially about the central axis of the head **110**.

(62) 3.4.5 Stacked Control Rings

(63) The control rings on the head **110** can be stacked on parallel planes, such that all control rings are operated by rotating the respective control rings about a shared central axis (e.g., the central axis of the head **110**). The stacked configuration allows a user to operate all controls using one hand, and creates a compact and robust form factor. The control rings can each have unique outer textures (e.g., splining, knurling, etc.), such that a user may discern each control ring by touch/feel alone.

(64) To maintain a small form factor and small effective diameter, the head **110** can be free of screw-knobs or hand-knobs. Moreover, each control ring can be fully engaged or disengaged by a single turn (or less), such that a user may lock or unlock all control rings with a single motion.

(65) In one implementation, the head **110** includes the set of stacked control rings including the camera lock ring **126**, the panning control ring **128**, and the pivot control ring **124**. The head **110** includes an upper body coupled to the base section **112** and rotatable about a pan axis of the base section **112**. In this implementation, the panning control ring **128** is arranged in between the camera lock ring **126** located on an upper section of the head **110** and the pivot control ring **124** located on the base section **112**. Further, the panning control ring **128** can be configured to lock the upper body of the head **110** to the base section **112** of the head **110** responsive to rotation in the first direction about the base section **112** and unlock the upper body from the base section **112** responsive to rotation in the second direction. Thus, when the upper body is unlocked from the base section, a user may continue operating each control ring as the camera lock ring **126** is located on the upper section of the head **110** and interacts with other components (e.g., locking tab **132**, rail

134) on the upper section and the pivot control ring **124** is located on the base section **112** and interacts with components (e.g., hat **116**, sun gear **120**, planet gears **122**) on the base section **112** and extending below.

(66) 3.5 Base Section

(67) A second side of the base section **112** includes a set of flanges **114** extending downward from the head **110**, which form an exposed spherical socket configured to receive and hold the spherical end **156**.

(68) In one implementation, the spherical socket includes three flanges **114** spaced at 120 degrees around the central axis of the head **110**. The flanges **114** can be configured to fit (e.g., nest) between the leg mounts **144** of the hub **140** section when the tripod **100** is in a full or partially-collapsed state for vertical packing efficiency. Each flange includes a concave surface on a side facing the inner socket area. A socket bushing can sit between the flanges **114** and the spherical end **156**. When the pivot control ring **124** is engaged, the reaction forces on the inner surfaces of the flanges **114** engage with the spherical socket bushing, which locks the spherical end **156** in a fixed position.

(69) In one implementation, the base section **112** includes a set of flanges **114**, each flange defining a pliable tip in contact with the spherical end **156**. The spherical end **156** can include a base material (e.g., an aluminum base material) and a surface coating deposited over the base material, such that pliable tips of the flanges **114** contact the surface coating of the spherical end **156**. The spring **117** of the head **110** can be preloaded to clutch the spherical end **156** between the hat **116** and pliable tips of the set of flanges. In this implementation, the spring rate and preload of the spring **117** can be matched to the surface finish of the spherical end **156** and a coefficient of friction of the pliable tips of the flanges **114** such as to retain an orientation of the head **110** on the spherical end **156** during rotation of the pivot control ring **124** in the first direction.

(70) For example, the base section **112** can include the set of flanges **114**, each flange including a rubber tip in contact with the spherical end **156** and exhibiting a coefficient of friction. The spherical end **156** can be constructed to include an aluminum base material and a surface coating deposited over the aluminum base material. The spring **117** can be preloaded according to the coefficient of friction of the rubber tips of the flanges **114** and the surface finish of the spherical end **156**.

4. Hub

(71) As shown in FIG. **10**, the hub **140** includes: a central shaft (e.g., center bore **142**) configured to slidably receive the center column **150**; lobes **146** extending outward from the central shaft and including a locking assembly configured to interface with the center column **150**; and leg mounts arranged in a radial pattern about the center bore **142** and **144** spaced between each adjacent pair of lobes **146**, the leg mounts **144** configured to interface with leg hinge-joints.

(72) Subsections of the leg mounts **144**, the lobes **146**, and the central shaft can combine to form a substantially hemispherical recess configured to receive a lower section of the spherical end **156** such that, in a fully collapsed state, the flanges **114** of the head **110** and the leg mounts **144** of the hub **140** and leg sections encapsulate the spherical end **156**. By configuring the spherical end **156** to nest within the head **110** and hub **140**, the tripod **100** exhibits increased vertical and volumetric efficiency and minimizes negative space.

(73) In one implementation, the hub **140** section includes a set of magnets configured to interact with magnetic features of each other section of the tripod (e.g., the head **110**, the legs **160**), such that the tripod **100** maintains a collapsed state in the absence of user interaction.

(74) 4.1 Center Bore

(75) The hub **140** defines a center bore **142** of the tripod **100**. The center bore **142** of the hub **140** can be configured to receive the center column **150** as well as lock the center column **150** in place. Generally, the center bore **142** defines a non-circular cross-section, thus preventing the center column **150** from rotating within the center bore **142**. The center bore **142** can include a shaft

bushing (e.g., a rubber or bronze bushing) in order to limit wear on the center column **150** resulting from extension and retraction of the center column **150** in the hub **140** over time.

(76) In one implementation, the center bore **142** defines a hexagonal cross section having irregular sides such that three non-adjacent faces of the center bore **142** each form the inner face of a hub lobe, and the remaining three non-adjacent faces of the center bore **142** each form the inner face of the base of each leg mount section.

(77) 4.2 Lobes of the Hub

(78) The hub **140** includes a set of lobes **146** extending outward from the central shaft. Each lobe can include an inner space to hold either a primary or secondary lock assembly configured to retain the center column **150** in a fixed or semi-fixed state. In one implementation, the space between each pair of lobes **146** is configured to nest a leg of the tripod **100**.

(79) 4.2.1 Center Column Lock Assembly

(80) A first lobe of the hub **140** can include the primary lock assembly. The primary lock assembly can include a cambered rocker arm placed within the first lobe and configured to apply force to the center column **150** when engaged by a threaded hand-screw. The rocker arm can be pinned at a bottom end of the rocker bar such that, when a hand-screw applies force at a top end of the rocker bar, the camber of the rocker bar (in connection with the top and bottom force points) creates an area of contact at a center area of the rocker bar. The camber of the rocker arm allows the rocker arm to disperse the force applied to the center column **150**. Thus, a thin-walled center column can sufficiently support forces applied to the center column **150**.

(81) A ball-detent lock assembly can be disposed in a second lobe of the hub **140**. The ball-detent lock assembly applies force in a first hub plane orthogonal to the central axis of the tripod **100** to hold the center column **150** in a temporarily fixed position. Each lobe of the hub **140** can include a ball-detent lock assembly. While the spring-loaded ball is in a position outside of the detent locations, the ball continues to apply force to the center column **150**.

(82) In one variation, the primary lock assembly includes a knob **148** configured to engage the cambered rocker arm. The knob **148** can be configured to extend in order to enable accessibility and easier adjustment when the tripod **100** is deployed and retract (e.g., nest between two legs) when the tripod **100** is collapsed or stored. For example, the knob **148** can include: a screw defining a threaded end and a splined bore; a shaft (e.g., a steel shaft) defining a first end press-fit into a cap and a second splined end configured to run inside the splined bore of the screw and to transiently couple to a magnetic element within the splined bore; and a spring configured to disengage the shaft from the magnetic element responsive to a user applying a force (e.g., pulling) on the cap in a direction opposite the magnetic element. In a collapsed state (e.g., when the tripod **100** is in storage), the second end of the shaft magnetically couples to and is retained inside of the splined bore by the magnetic element within the splined bore. Thus, in this collapsed state, the cap can nest between two adjacent legs **160** and therefore reduce a cross-section and an effective maximum diameter of the tripod **100**. However, when a user pulls on the cap and overcomes magnetically coupling between the shaft and the magnetic element, the shaft disengages from the magnetic element and moves outwardly from the splined bore, and the spring retains the shaft in this extended state. In this extended state, the cap of the knob **148** is offset outwardly from the two adjacent legs, thereby enabling greater access to the knob **148** and easier adjustment of the center column position for the user. To return the cap to the retracted state, the user may depress the cap, thereby overcoming the spring and re-coupling the shaft to the magnetic element.

(83) 4.3 Leg Mounts

(84) Generally, the leg mounts **144** are configured to connect each leg of the leg section to the hub **140** at a hinge joint. The leg mounts **144** are also configured such that the flanges **114** of the spherical socket fit between the leg mounts **144** when the center column **150** is fully depressed into a collapsed state.

(85) In one implementation, the leg mounts **144** include a multistage position stop (or “stop”), such

that each leg can lock in at least a first position and a second position. For example, the stop can enable legs of the tripod **100** to operate in a set of positions including an open position defined by legs offset from the central axis and extending outwardly from the hub **140** at a first angle of 25-degrees (+/-2 degrees), a low position defined by legs offset from the central axis and extending outwardly from the hub **140** at a second angle between 75 and 85-degrees (+/-2 degrees), and a collapsed position defined by legs approximately parallel to the central axis.

(86) 4.4. Packed Configuration

(87) The leg mounts **144** extend from the hub **140** and are arranged in a radial configuration about the center axis (e.g., at 0-degree, 120-degree, and 240-degree intervals). Furthermore, interior faces of the leg hub **140** mounts are relieved to enable the spherical end **156** to nest in hub **140**—that is, the interior faces of the leg hub **140** mounts are relieved to enable the spherical end **156** to drop into the hub **140** and to be encapsulated within the leg mounts **144**. The hub **140** also defines gaps (or “opens”) between adjacent ends of adjacent leg mounts **144**, and the flanges **114**—extending downwardly from the head **110** and spaced radially about the central axis of the head **110** (e.g., at 0-degree, 120-degree, and 240-degree intervals, like the leg mounts **144**)—define widths (slightly) less than the gap width between adjacent leg mounts **144** such that these flanges **114** can nest in these gaps between leg mounts **144** when the tripod **100** is collapsed, thereby limiting total height and increasing volumetric efficiency of the collapsed tripod **100**.

(88) Furthermore, because the leg mounts **144** are relieved for the spherical end **156**, the spherical end **156** can define a relatively large diameter, thereby enabling the flanges **114** and the hat **116** to cooperate to apply a relatively large clamping force to the spherical end **156** and thus support relatively large cantilevered masses arranged on the head **110** (e.g., a large telephoto lens installed on a camera mounted to the head **110**) without increasing the height or reducing volumetric efficiency of the tripod **100** when collapsed. For example, the diameter of the spherical end **156** can be greater than a minimum distance from the top faces of the hub lobes **146** to the bottom face of the pivot control ring **124** when the tripod **100** is collapsed.

(89) Furthermore, the legs **160** can include magnetic and/or ferrous elements arranged proximal their distal ends and configured to attract magnetic and/or ferrous elements in adjacent legs **160** when the tripod **100** is collapsed, thereby retaining these distal ends of the legs **160** in close proximity and preventing inadvertent expansion of the legs **160** during transport.

5. Center Column

(90) The center column **150** can be configured to translate within the center bore **142** of the hub **140**. The center column **150** can have a non-circular cross-section to prevent rotation of the center column **150** within the center bore **142**. In one implementation, the center column **150** defines a tri-lobed cross-section. In this implementation, the center bore **142** defines a tri-lobed opening with lobes **146** radially centered between leg mounts **144** of the hub **140**.

(91) In another implementation, as shown in FIG. 4, the center column **150** can be segmented into a set of center column **150** modules. In this implementation, the center column **150** includes a center column stub **152** and a center column extension **154**, wherein the center column stub **152** can be formed from a different material than the center column extension **154**. The center column stub **152** can be attached or detached from the center column extension **154** via a fastener located within an access point inside the spherical end **156**, accessible when the head **110** is actuated to a full 90-degree configuration. Moreover, modular instances of the center column **150** can be added to expand the total height of the tripod **100**.

(92) The center column stub **152** can function as a center column **150**. In one implementation, the center column stub **152** can be of sufficient height such that the center column stub **152** a full range of motion for the head **110**. The center column stub **152** can be separated from the center column extension **154** via a fastener within an access point in the spherical end **156**, the access point accessible between the spherical socket flanges **114** when the main plane of the head **110** is in a 90-degree orientation with respect to the main axis of the tripod **100** (i.e. the main axis of the center

column **150**).

(93) Furthermore, when legs **160** are deployed during operation but the center column **150** remains retracted, the head **110** can remain nested in the hub **140** section such that the hub **140** section mechanically engages and retains the head **110**, thereby enabling the head **110** to support a large cantilevered load (e.g., a telephoto lens) rather than rely on friction between the spherical socket flanges **114**, hat **116**, and spherical end **156** to support this load.

(94) The center column **150** can be constructed from a strong and durable material such that the center column **150** supports a minimum load. In one implementation, the center column **150** is constructed from aluminum.

(95) 5.1 Spherical End

(96) The spherical end **156** can connect to a first end of the center column **150**. Generally, the spherical end **156** can be housed in a socket of the head **110**, such that the head **110** can pivot about the spherical end **156**. In one implementation, the spherical end **156** is coupled to an end of the center column **150** opposite the set of legs **160** and is configured to nest between the leg mounts **144** of the hub **140**.

(97) In this implementation, the spherical end **156** can be configured to nest between the leg mounts **144** such that the spherical center of the spherical end **156** falls on or near a horizontal “pivot plane” intersecting the pivot axes of the legs **160**, such as less than the spherical radius of the spherical end **156** from the pivot plane. Similarly, the spherical end **156** can be configured to nest between the leg mounts **144** such that a bottom of the spherical end **156** falls below the pivot plane and such that bottoms of the flanges **114** fall below the pivot plane when the head **110** is fully collapsed into the hub **140**.

(98) The spherical end **156** can also include a scratch-resistant outer coating. In one implementation, the spherical end **156** is constructed from an aluminum base material and includes a scratch-resistant (e.g., rubberized or hard-anodize) coating over the aluminum base material.

(99) 5.2 Hanging Hook

(100) A hanging hook **158** can connect to a second end of the center column **150**, such that a user may hang a bag or weight from the hanging hook **158** for additional stability. Generally, the hanging hook **158** includes: a first projection having a first cross section including a profile matching an inner cross-section of the center column **150**; a retractable second projection having a second cross section matching an outer cross-section of the center column **150**; and a hook. The first projection can include a set of bosses configured to fit a set of detents on the inner walls of the center column **150**. While retracted, the second projection allows for turning the first projection inside the center column **150**, such that the set of bosses can access the set of detents. When not retracted, the second projection restricts rotation of the hanging hook **158** within the center column **150** by filling the (non-circular) interior cross-section of the center column **150**.

(101) In one implementation, the hanging hook **158** can also function as a hard stop for the center column **150**, thereby preventing a user from unintentionally withdrawing the center column **150** fully out of the center bore **142** when raising the center shaft to a maximum height above the hub **140**. For example, the hanging hook **158** can include a first end defining a hook configured to carry a weighted body and a second end opposite the hook and configured to attach to a distal end of the center column opposite the head to prevent passage of the distal end through the center bore of the hub. Thus, to release the center column **150** from the hub **140**, the user may first remove the hanging hook **158** from the bottom end of the center column **150**. (After removing the center column **150** from the hub **140**, the user may also retrieve a mobile mount **180** from inside the center column **150**, as described below.)

(102) In one variation, as shown in FIG. 6, the hanging hook **158** can include: a first end defining a hook; and a second end opposite the hook and including a magnetic element configured to couple to a corresponding magnetic feature or ferrous element integrated into an end of a mobile mount **180**—described below—stored inside the center column **150**. In this variation, when the hanging

hook **158** is locked into the center column **150**, the hanging hook **158** can cooperate with a spring element located within the center column **150**—offset above the hanging hook **158**—to constrain the mobile mount **180** within the center column **150**. Furthermore, when the mobile mount **180** is partially ejected from the bottom of the center column **150** but retained by the spring element, the magnetic element in the hanging hook **158** can couple to the magnetic or ferrous element in the mobile mount **180** in order to coaxially align the hanging hook **158** to the mobile mount **180** and the bore of the center column **150**, thereby providing positive feedback to the user as the user inserts the hanging hook **158** into the center column **150**.

(103) Furthermore, when the tripod **100** is fully retracted, the center column **150** can locate the hanging hook **158** near feet at ends of the legs **160** such that the hook is physically accessible when the tripod **100** is fully retracted, thereby enabling a user to hook the tripod **100** directly to a bag (e.g., a camera or equipment bag), belt loop, or other hoop for transport.

(104) 5.3 Mobile Mount

(105) As shown in FIGS. 6, 7A, and 7B, the tripod **100** can also include a collapsible mobile phone mount **180** (hereinafter “mobile mount”) arranged within the center column **150**. Generally, the mobile mount **180** can be configured to receive and hold a mobile phone in an open position. The mobile mount **180** can be configured to transiently attach to the camera platform **130**. The mobile mount **180** can collapse down to a diameter less than the diameter of the center column **150** in a closed position. In one variation, the mobile mount **180** is spring-loaded and magnetically attached within the center column **150**, such that—upon removal of the hanging hook **158** at the end of the center column **150**—the mobile mount **180** ejects itself from the center column **150** and expands into a deployed configuration for a user to clamp a mobile phone into, and then fix the mobile mount **180** onto the camera platform **130** of the tripod **100**. Therefore, the center column **150** can define a cavity, opposite the spherical end **156**, configured to house the mobile mount **180** in the collapsed state.

(106) In one implementation, the center column **150**: defines a distal end configured to receive the hanging hook **158** described above; and includes a spring loaded detent with a magnetic element offset above the distal end of the center column **150**. In this implementation, the spring loaded detent can be offset above the distal end by less than a collapsed length of the mobile mount **180** such that the spring loaded detent retains the mobile mount **180** in the center column **150** with a portion (e.g., approximately ten millimeters) of the opposite end of the collapsed mobile mount **180** extending out of the distal end of the center column **150**, thereby enabling a user to grasp and withdraw the mobile mount **180** from the center column **150** when the hanging hook **158** is removed from the center column **150** as shown in FIG. 6. However, when the hanging hook **158** is offered up to an end of the mobile mount **180** hanging out of the center column **150** and lifted up into the center column **150** by a user, the spring loaded detent can compress to accommodate insertion of the mobile mount **180** and the hanging hook **158** into the bore of the center column **150**.

(107) Therefore, the mobile mount **180** can include: a first magnetic feature configured to magnetically couple to the spring loaded detent inside the center column **150**; and a second magnetic feature—opposite the first magnetic feature—configured to mate with a ferrous component located in the hanging hook **158**.

(108) For example, when the mobile mount **180** is collapsed, the mobile mount **180** can define a first end including a first magnetic feature and a second end including a second magnetic feature. The first magnetic feature can mate with a magnetic element located within the center column **150**, the magnetic element configured to retain the mobile mount **180** within the center column **150**. The second magnetic feature can mate with a ferrous component of the hanging hook **158**, such that the hanging hook **158** can first connect (e.g., magnetically) to the mobile mount **180** when reattaching the hanging hook **158** to the tripod **100**. Additionally, the mobile mount **180** can engage with the spring loaded detent located within the center column **150** such that, when the hanging hook **158** is

attached to the center column **150**, the mobile mount **180** is fully inserted within the center column **150** and the spring **117** is compressed. Then, when the hanging hook **158** is removed from the center column **150** (e.g., by the user), the mobile mount **180** can disengage from the spring loaded detent and drop down (e.g., drop one inch) within the center column **150** before the first magnetic feature of the mobile mount **180** engages the magnetic element in the center column **150**. Therefore, when the hook is removed, the mobile mount **180** can drop slightly within the center column **150**—without falling out of the center column **150** completely—such that a user may easily remove the mobile mount **180** from the center column **150**.

(109) Once removed from the bore of the center column **150**, the mobile mount **180** can attach to the camera platform **130** to enable a user to mount a mobile device (e.g., a smartphone) to the tripod **100**, as shown in FIG. 7A. For example, a user may: remove the hanging hook **158** from the center column **150**; withdraw the mobile mount **180** from the center column **150** in the collapsed position; expand the mobile mount **180** to the open position to retain sides of a mobile device; locate the mobile mount **180** on the camera platform **130**; and then rotate the camera lock ring **126** to lock the mobile mount **180** to the camera platform **130**. After shooting with the mobile device, the user may: remove her mobile device from the mobile mount **180**, which releases the mobile mount **180** to automatically return to the collapsed state; rotate the camera lock ring **126** to unlock the mobile mount **180** from the camera platform **130**; remove the mobile mount **180** from the camera platform **130**; insert the mobile mount **180** back into the center column **150**; and replace the hanging hook **158** at the distal end of the center column **150**.

(110) 5.4 Center Column Geometry

(111) In one implementation, the center column **150** defines a tri-lobed cross-section, with each lobe **200** radially centered between two adjacent legs **160** extending from the leg mount. In this implementation, recessed faces of the center column **150**—between adjacent lobes **200**—provide clearance for legs **160** of the tripod **100** to collapse more closely and enable the tripod **100** to reduce to a smaller maximum width when fully collapsed, as shown in FIG. 1. Additionally, lobes **200** of the tripod **100**—radially offset by 120° about the center column **150**—yield a larger effective moment of inertia and thus yield less deflection and vibration under greater load (e.g., a large camera and/or lens loaded onto the camera platform **130**) and at greater extension above the hub leg mount **144**. More specifically, this tri-lobed center column **150** defines three recessed faces **201**—radially offset by 120°—and yields greater clearance along inside faces of the legs **160**, thereby enabling the legs **160** to pack into a smaller volume when fully retracted and closed. Additionally, the tri-lobed center column **150** exhibits greater effective moment of inertia than a round or hexagonal column of the same dimension between recess faces **201**, thereby enabling the center column **150** to carry greater loads at greater heights above the hub leg mount **144** with less deflection and lower vibration amplitude.

(112) In another implementation, the center column **150** defines a cross-section including a number of sides equal to double a number of lobes **200** between adjacent legs extending from the leg mount. For example, the center column **150** can define a cross section of an irregular hexagon with a first set of three sides, each having a first length, and a second set of three sides, each having a second length. In this implementation the spherical end **156** includes three flanges **114**, and the hub **140** includes three lobes **146**. The central column can be dynamically locked in place by a screw locking mechanism that screws into threads along an axis orthogonal to the central main axis.

6. Legs

(113) Each leg **160** includes leg sections **162** configured to nest within an adjacent leg section **162** by sliding along a shared axis. Smaller leg sections **162** can be locked in place by a set of leg section locks **172** (or “clamp assemblies”). The leg section locks **172** are activated by flip locks (e.g., c-clamps) that abut each leg joint. Generally, the leg section locks **172** define a height significantly shorter than a height of a leg section **162**. In one implementation each leg **160** includes five distinct leg sections **162**.

(114) Each leg can splay outward from a central vertical axis up to an angle defined by a multistage leg position stop (or “stop”). Each leg is configured to splay further up to at least a second angle defined by the leg lock assembly responsive to actuation of the multistage leg position stop.

(115) In one implementation, each leg includes a shaft with six faces, three inward-facing and three outward-facing, such that when the tripod **100** is in a fully collapsed state, each of the inward faces of each leg sit parallel with an inward face of an adjacent leg or a face **201** of the central column.

(116) Furthermore, because each leg defines a width (e.g., spans an arc length about the center axis) greater than its depth, each leg of the tripod **100** can thus exhibit a greater area moment of inertia in its bending axis and less deflection when subject to a yaw load than a round leg.

Therefore, the legs **160** can cooperate to resist deflection and minimize vibration in yaw as a user rotates a camera—loaded onto the head **110**—such as when shooting a video pan of a car drive-by.

(117) 6.1 Leg Assembly and Lightweight Mode

(118) In one variation, as shown in FIG. **9**, lower telescoping leg sections are removable from the uppermost leg section and are replaceable with a foot **164** insert for each leg of the tripod **100** in order to reduce overall weight of tripod **100**, such as when a user is backpacking or otherwise desires reduced pack weight.

(119) In one implementation, as shown in FIGS. **8A** and **8B**, the first leg of the tripod **100** includes a first, uppermost leg section defining: a proximal end pivotably coupled to the hub leg mount **144**; a distal end defining a notch extending circumferentially about one lateral side of the first leg section; and a distal end including a perforation, dimple, or other engagement feature opposite the notch and configured to retain a foot **164**, as described below.

(120) In this implementation, the first leg of the tripod **100** further includes an upper clamp assembly **170**. The upper clamp assembly **170** includes a c-clamp body defining: a longitudinal split extending along the full height of the c-clamp body; a clamp bore of internal cross-section approximating the outer cross-section of the distal end of the first leg section (e.g., ± 1 millimeter); a lower clamp flange adjacent a first side of longitudinal split; an upper clamp flange adjacent the first side of the longitudinal split and above the lower clamp flange; a lower clamp surface adjacent a second side of the longitudinal split and facing the lower clamp flange; and an upper clamp surface adjacent the second side of the longitudinal split and facing the upper clamp flange. The upper clamp assembly also includes a leg bushing: arranged inside the clamp bore proximal a bottom of the c-clamp body to fill a gap between the clamp bore and an outer surface of a second leg section running inside the clamp bore; and including a flange configured to insert into the distal end of first leg section and to fill a gap between the internal bore of the first leg section and the outer surface of the second leg section running inside the first leg section.

(121) In this implementation, the upper clamp assembly **170** further includes a lower clamp: pivoting transiently in the lower clamp surface; coupled to the lower clamp flange; configured to draw the lower clamp flange toward the lower clamp surface in a closed position in order to compress the c-clamp body around the second leg section running inside the c-clamp body and thus lock the upper clamp assembly **170** to the second leg section; and configured to release the lower clamp flange from the lower clamp surface in an open position in order to release the c-clamp body from the second leg section and thus enable the second leg section to telescope within the first leg section. Additionally, the upper clamp assembly **170** includes: an upper clamp pivoting transiently in the upper clamp surface; coupled to the upper clamp flange; configured to draw the upper clamp flange toward the upper clamp surface in a closed position in order to compress the c-clamp body around the distal end of the first leg section and thus lock the upper clamp assembly **170** to the first leg section; and configured to release the upper clamp flange from the upper clamp surface in an open position in order to release the c-clamp body from the first leg section and enable removal of the upper clamp assembly **170**, all lower leg sections, and all lower clamp assemblies from the first leg section.

(122) Furthermore, in this implementation, the c-clamp body defines a lateral split extending

laterally from both sides of the longitudinal split between the upper and lower clamp flanges **114**. The c-clamp body locates the lateral split adjacent the notch extending circumferentially about one lateral side of the distal end of the first leg section, thereby isolating compression of the c-clamp body—by the upper clamp in the closed position—onto the distal end of the first leg section rather than onto the second leg segment below, and similarly, isolating compression of the c-clamp body—by the lower clamp in the closed position—onto the proximal end of the second leg section rather than onto the first leg segment above.

(123) The first leg can include additional leg sections (e.g., second leg section, third leg section, etc.) with c-clamp assemblies (e.g., leg locks **172**) interposed between these lower leg sections, as shown in FIGS. **5**, **8A**, and **8B**. Additionally, each other leg in the tripod **100** can include leg sections of similar geometry and can include similar upper clamp assemblies.

(124) As shown in FIGS. **9A** and **9B**, the tripod **100** can further include a set of feet. In this variation, a foot includes a proximal end configured to insert into the distal end of an upper leg section—in a particular leg of the tripod **100**—when the upper clamp assembly **170** and lower leg sections are removed from the upper leg section of this particular leg. The proximal end of the foot includes a detent configured to engage a perforation, dimple, or other feature defined at the distal end of the upper leg section in order to transiently retain the foot to this leg section. Additionally, the foot includes a foot surface extending longitudinally from its proximal end.

(125) Therefore, for full height range adjustment in a full-assembly mode, the user: installs lower leg sections and upper clamp assemblies into each first leg section; and installs the center column extension **154** onto the center column stub **152**. To reduce weight and maintain some height range adjustment in a lightweight mode, the user: removes lower leg sections and upper clamp assemblies from each first leg section; installs a foot into the distal end of each first leg section; and retains the center column extension **154** on the center column stub **152**. To minimize weight in a full-lightweight mode, the user: removes lower leg sections and upper clamp assemblies from each first leg section; installs a foot into the distal end of each first leg section; and removes the center column extension **154** from the center column **150**. However, in the full-lightweight mode, the tripod **100** can still enable some height adjustment. For example, the user may install the center column stub **152** in the center bore **142** of the hub **140** with the controls chassis either extending above the hub **140** (e.g., with a camera upright) or below the hub leg mount **144** (e.g., with the camera inverted).

(126) As a person skilled in the art will recognize from the previous detailed description and from the figures and claims, modifications and changes can be made to the embodiments of the invention without departing from the scope of this invention as defined in the following claims.

Claims

1. A tripod comprising: a hub defining a center bore; a set of legs, each leg in the set of legs having an interior face and being pivotably or retractably coupled to the hub; a center column comprising a first non-circular cross-section with exterior faces, said center column configured to translate within the center bore; and a head coupled to the center column configured to couple to a camera, wherein the exterior faces of the first non-circular cross-section of the center column and the interior faces of the legs are configured to mate or fall very near thereby minimizing negative space between the set of legs and the center column when the tripod is in a collapsed state.
2. The tripod of claim 1, wherein the center column comprises a triangular cross-section.
3. The tripod of claim 1 where the center column has a hexagonal cross-section.
4. The tripod of claim 3 wherein the hexagonal cross-section has irregular sides with a first set of three nonadjacent sides each having a first length and a second set of three nonadjacent sides each having a second length.
5. The tripod of claim 1 wherein the center bore defines a second non-circular cross section

configured to prevent rotation of the center column within the center bore.

6. The tripod of claim 5 wherein the center column comprising the first non-circular cross section nests within the second noncircular cross-section of the center bore.
